Analytical Chemistry Cumulative Examination
December 02, 2022
Topic: Electrochemistry

1. Electrochemistry Basics.

The following reaction is the **overall chemical reaction** occurring in a galvanic (spontaneous) electrochemical cell involving oxygen (O_2) at a gold electrode in an electrolyte solution with pH adjusted (H^+ concentration controlled) and a silver electrode in a solution containing Br^-. (Note that the O2-containing part of the cell is saturated with O_2.)

$$\frac{1}{2}\,O_2 + 2\,Ag(s) + 2\,Br^- + 2\,H^+ = 1\,H_2O + 2\,AgBr(s)$$

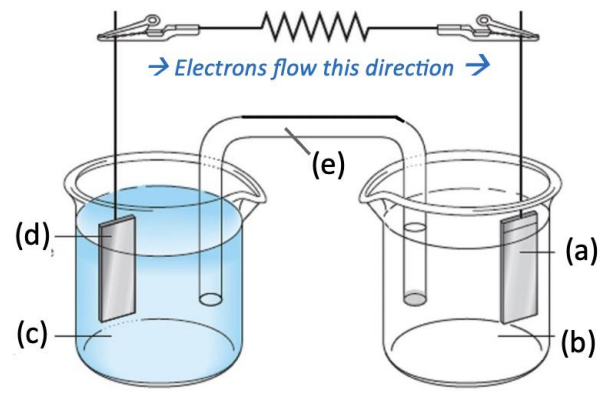
The diagram of the electrochemical cell shows which direction the electrons are going in the external circuit.

From the table of standard electrode potentials on **page 7** in the *Equations/Constants/Data* document, the **two half reactions** that are involved in this reaction and their standard reduction potentials are:

$AgBr(s) + e^- = Ag(s) + Br^-(aq)$ ***E° = 0.0711 V vs. NHE***

$(1/2)\,O_2 + 2\,H^+ + 2\,e^- = H_2O\,(l)$ ***E° = 1.229 V vs. NHE***

a. (2.5 points) Match the letters (a), (b), (c), (d), and (e) from the diagram to each of the following terms. Note that the diagram already shows you which direction the electrons are going:
(Use only one letter per term and each letter can only be used once).



(i) Anode _____

(ii) Cathode _____

(iii) Salt bridge _____

(iv) Electrolyte solution containing Br- _____

(v) Electrolyte solution containing H^+ _____

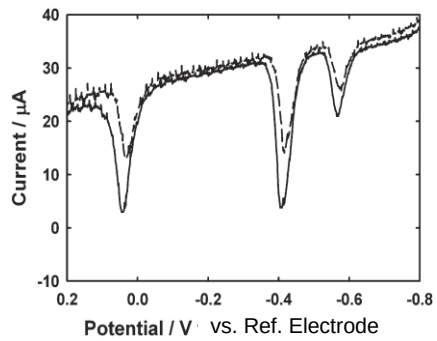
b. (10 points) What is E_{rxn} under standard conditions everywhere, except the pH of the solution on the oxygen side is made neutral, pH = 7, $[H^+] = 1.0 \times 10^{-7}$ M. (Hint: Use the Nernst equation for both half reactions to determine the E_{rxn} under these non-standard conditions. Remember that solids have activities of unity (concentration term for solids is 1).

2. Anodic Stripping Voltammetry.

A new approach to enhance the detection sensitivity of anodic stripping voltammetry on a small volume scale (150 µL) was developed where the deposition step involves stirring for only 60 s using magnetohydrodynamics (all that is required is a magnetic field during the deposition step and no stir bar!). The stripping response is shown below in the presence (solid curve) and absence (dashed curve) of the magnetic field. The reference electrode (Ref. Electrode A) used in this experiment has a potential of approximately +0.260 V vs. NHE. *Identify the metals in the mixture using the following table of species where the E° values are given vs. NHE. Show calculations to determine which element can be assigned to each peak.*

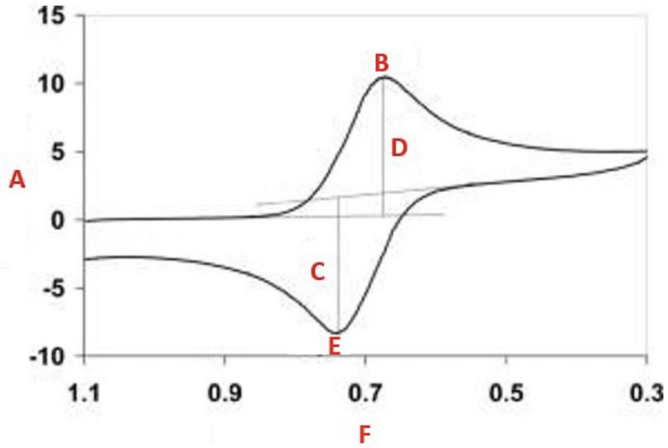
E° values vs. NHE

Ag+ + e = Ag°	+0.7991
Cu^{2+} + 2e = Cu°	+0.340
Pb^{2+} + 2e = Pb°	-0.1205
Cd^{2+} + 2e = Cd°	-0.3515
Fe2+ + 2e = Fe°	-0.44
Zn^{2+} + 2e = Zn°	-0.7626



3. <u>Cyclic Voltammetry.</u>

Here is an example of a cyclic voltammetry (CV) response of a water-soluble inorganic complex called "O", that you recently synthesized. (Note: "O" means "oxidized form" here; it does ***not*** refer to oxygen. Also, "R: means "reduced form" here; it does not refer to the ideal gas constant).You want to know about this complex's electrochemical properties so that you can determine whether to use it in a new redox-flow battery.



Your cyclic voltammogram was obtained at 25 °C at a platinum disk working electrode having an area of 2.54 mm^2 and start the CV at +1.1 V vs. SCE, sweep to +0.3 V, and then sweep back to +1.1 V at 0.100 V/s scan rate. (The counter electrode is a typical platinum flag electrode.) The plot uses the "standard" or "Texas" graphing convention (NOT IUPAC). This was obtained for an **unknown concentration** of the inorganic complex "O" in a supporting electrolyte of 0.1 M KNO_3. The supporting electrolyte provides an ionically conducting solution and does not undergo an electron transfer event at the electrode (this is why it is called ***supporting*** electrolyte). It is known that the **diffusion coefficient for both forms of the redox couple is 6.00 x 10^{-6} cm^2/s**. Assume activity coefficients are unity.

The half reaction for the reduction of O to R is:
$O_{(aq)} + ne^- = R_{(aq)}$

(a) (2.5 points) Match the letters on the figure to the correct label from the list. (Note that not all options will be used).

(i) Potential (mV)

(ii) Current (µA)
(iii) Scan rate (*v* in V/s)
(iv) Cathodic peak potential (E_{pc})
(v) Anodic peak potential (E_{pa})
(vi) Concentration (mM)
(vii) Anodic peak current (i_{pa})
(viii) Cathodic peak current (i_{pc})
(ix) Gibbs free energy (J)

A._____
B._____
C._____
D._____
E._____
F._____

(b) (3 points) Given this CV response, determine the E° value for the half reaction?

(c) (3 points) Determine ΔE_p from the figure. What is the expected value of ΔE_p for a fast electron transfer reaction?

(d) (3 points) Determine the value of n. (How many electrons are transferred per molecule of the inorganic complex? Assume that the complex undergoes a fast electron transfer).

(e) (3 points) Consider the peak currents in the figure. Draw a conclusion about the stability of O and R in the solution.

(f) (3 points) From the figure and the information provided and obtained above in this question, calculate the concentration of the complex in the solution.

Electrochemistry Basics, Anodic Stripping Voltammetry

and Some Cyclic Voltammetry

Here, *R* can be either represent the reduced form of a redox couple, O + ne$^-$ ➔ R, or the ideal gas constant, R = 8.3144621(75) J K^{-1} mol^{-1}. Therefore, be careful how you interpret the equations.

E = electric potential

Using logarithm base e:

$$E = E^{o'} + \frac{RT}{nF}\ln\left(\frac{[O]}{[R]}\right) = E^{o'} + \frac{RT}{nF}\ln\left(\frac{\text{oxidized forms}}{\text{reduced forms}}\right)$$

$$E = E^{o'} + \frac{0.02569}{n}\ln\left(\frac{[O]}{[R]}\right), \text{ at 25 °C}$$

$E^{o'} = E^{o}$ when activity coefficients, $\gamma_O = \gamma_R = 1$

Using logarithm base 10:

$$E = E^{o'} + \frac{2.303RT}{nF}\log\left(\frac{[O]}{[R]}\right) = E^{o'} + \frac{2.303RT}{nF}\log\left(\frac{\text{oxidized forms}}{\text{reduced forms}}\right)$$

$$E = E^{o'} + \frac{0.05916}{n}\log\left(\frac{[O]}{[R]}\right) = E^{o'} + \frac{0.05916}{n}\log\left(\frac{\text{oxidized forms}}{\text{reduced forms}}\right), \text{ at 25 °C}$$

(E° = E°' when activity coefficients are unity)

$$\Delta G^{o}_{rxn} = -n\,F\,E^{o}_{rxn}$$

$$E^{o}_{rxn} = E^{o}_{cell} = E^{o}_{cathode} - E^{o}_{anode} = E^{o}_{reduction\ half\ reaction} - E^{o}_{oxidation\ half\ reaction}$$

E_X vs. new ref. = E_X vs. old ref. - $E_{\text{new ref.}}$ vs. old ref.

$i_p = (n^2 F^2 v\, l\, A\, C_M^*)/ (2.7\, R\, T)$

$$\upsilon = \text{scan rate} = \frac{\Delta E}{\Delta t} = \frac{\left(E_{final} - E_{initial}\right)}{\left(t_{final} - t_{initial}\right)}$$

$$E_{1/2} = \frac{E_{pf} + E_{pr}}{2}$$

$$E_{1/2} = E^{0'} + \frac{RT}{nF}\ln\left(\frac{D_R}{D_O}\right)^{1/2}$$

$E_{1/2} = E^{o}$ *when* $D_R = D_O$ *and activity coefficients are unity*

$$\Delta E_p = \left|E_{pf} - E_{pr}\right| \approx \frac{2.3RT}{nF} = \frac{59}{n} mV \text{ at } 25°\text{C} = \frac{58}{n} mV \text{ at } 25°\text{C repeated cycling}$$

$$i_{pf} = 0.4463\left(\frac{F^3}{RT}\right)^{1/2} n^{3/2} A D_O^{1/2} C_O^* v^{1/2}$$

$$i_{pf} = \left(2.69 \times 10^5\right) n^{3/2} A D_O^{1/2} C_O^* v^{1/2} \text{ at } 25°\text{C}$$

Diffusion length $\delta = (2\ D\ t)^{1/2}$

$\left|i_c\right| = C_{dl}^* A v$ (for charging current in a voltammogram)

$\left|i_p\right| = \frac{n^2 F^2 |\upsilon| l A C_M^*}{2.7RT}$ (for peak current in a linear sweep anodic stripping voltammogram at a mercury film electrode))

Ohm's law: $iR = V$

REFERENCE T.

TABLE C.1 Selected Standard Electrode Potentials in Aqueous Solutions at 25°C in V *vs.* NHE[a]

Reaction	Potential, V
$Ag^+ + e \rightleftarrows Ag$	0.7991
$AgBr + e \rightleftarrows Ag + Br^-$	0.0711
$AgCl + e \rightleftarrows Ag + Cl^-$	0.2223
$AgI + e \rightleftarrows Ag + I^-$	−0.1522
$Ag_2O + H_2O + 2e \rightleftarrows 2Ag + 2OH^-$	0.342
$Al^{3-} + 3e \rightleftarrows Al$	−1.676
$Au^+ + e \rightleftarrows Au$	1.83
$Au^{3+} + 2e \rightleftarrows Au^+$	1.36
p-benzoquinone + $2H^+ + 2e \rightleftarrows$ hydroquinone	0.6992
$Br_2(aq) + 2e \rightleftarrows 2Br^-$	1.0874
$Ca^{2+} + 2e \rightleftarrows Ca$	−2.84
$Cd^{2+} + 2e \rightleftarrows Cd$	−0.4025
$Cd^{2+} + 2e \rightleftarrows Cd(Hg)$	−0.3515
$Ce^{4+} + e \rightleftarrows Ce^{3+}$	1.72
$Cl_2(g) + 2e \rightleftarrows 2Cl^-$	1.3583
$HClO + H^+ + e \rightleftarrows \frac{1}{2}Cl_2 + H_2O$	1.630
$Co^{2+} + 2e \rightleftarrows Co$	−0.277
$Co^{3+} + e \rightleftarrows Co^{2+}$	1.92
$Cr^{2+} + 2e \rightleftarrows Cr$	−0.90
$Cr^{3+} + e \rightleftarrows Cr^{2+}$	−0.424
$Cr_2O_7^{2-} + 14H^+ + 6e \rightleftarrows 2Cr^{3+} + 7H_2O$	1.36
$Cu^+ + e \rightleftarrows Cu$	0.520
$Cu^{2+} + 2CN^- + e \rightleftarrows Cu(CN)_2^-$	1.12
$Cu^{2+} + e \rightleftarrows Cu^+$	0.159
$Cu^{2+} + 2e \rightleftarrows Cu$	0.340
$Cu^{2+} + 2e \rightleftarrows Cu(Hg)$	0.345
$Eu^{3+} + e \rightleftarrows Eu^{2+}$	−0.35
$1/2F_2 + H^+ + e \rightleftarrows HF$	3.053
$Fe^{2+} + 2e \rightleftarrows Fe$	−0.44
$Fe^{3+} + e \rightleftarrows Fe^{2+}$	0.771
$Fe(CN)_6^{3-} + e \rightleftarrows Fe(CN)_6^{4-}$	0.3610

(*continued*)

TABLE C.1 (*continued*)

Reaction	Potential,
$2H^+ + 2e \rightleftarrows H_2$	0.0000
$2H_2O + 2e \rightleftarrows H_2 + 2OH^-$	−0.828
$H_2O_2 + 2H^+ + 2e \rightleftarrows 2H_2O$	1.763
$2Hg^{2+} + 2e \rightleftarrows Hg_2^{2+}$	0.9110
$Hg_2^{2+} + 2e \rightleftarrows 2Hg$	0.7960
$Hg_2Cl_2 + 2e \rightleftarrows 2Hg + 2Cl^-$	0.26816
$Hg_2Cl_2 + 2e \rightleftarrows 2Hg + 2Cl^-$ (sat'd. KCl)	0.2415
$HgO + H_2O + 2e \rightleftarrows Hg + 2OH^-$	0.0977
$Hg_2SO_4 + 2e \rightleftarrows 2Hg + SO_4^{2-}$	0.613
$I_2 + 2e \rightleftarrows 2I^-$	0.5355
$I_3^- + 2e \rightleftarrows 3I^-$	0.536
$K^+ + e \rightleftarrows K$	−2.925
$Li^+ + e \rightleftarrows Li$	−3.045
$Mg^{2+} + 2e \rightleftarrows Mg$	−2.356
$Mn^{2+} + 2e \rightleftarrows Mn$	−1.18
$Mn^{3+} + e \rightleftarrows Mn^{2+}$	1.5
$MnO_2 + 4H^+ + 2e \rightleftarrows Mn^{2+} + 2H_2O$	1.23
$MnO_4^- + 8H^+ + 5e \rightleftarrows Mn^{2+} + 4H_2O$	1.51
$Na^+ + e \rightleftarrows Na$	−2.714
$Ni^{2+} + 2e \rightleftarrows Ni$	−0.257
$Ni(OH)_2 + 2e \rightleftarrows Ni + 2OH^-$	−0.72
$O_2 + 2H^+ + 2e \rightleftarrows H_2O_2$	0.695
$O_2 + 4H^+ + 4e \rightleftarrows 2H_2O$	1.229
$O_2 + 2H_2O + 4e \rightleftarrows 4OH^-$	0.401
$O_3 + 2H^+ + 2e \rightleftarrows O_2 + H_2O$	2.075
$Pb^{2+} + 2e \rightleftarrows Pb$	−0.1251
$Pb^{2+} + 2e \rightleftarrows Pb(Hg)$	−0.1205
$PbO_2 + 4H^+ + 2e \rightleftarrows Pb^{2+} + 2H_2O$	1.468
$PbO_2 + SO_4^{2-} + 4H^+ + 2e \rightleftarrows PbSO_4 + 2H_2O$	1.698
$PbSO_4 + 2e \rightleftarrows Pb + SO_4^{2-}$	−0.3505
$Pd^{2+} + 2e \rightleftarrows Pd$	0.915
$Pt^{2+} + 2e \rightleftarrows Pt$	1.188
$PtCl_4^{2-} + 2e \rightleftarrows Pt + 4Cl^-$	0.758
$PtCl_6^{2-} + 2e \rightleftarrows PtCl_4^{2-} + 2Cl^-$	0.726
$Ru(NH_3)_6^{3+} + e \rightleftarrows Ru(NH_3)_6^{2+}$	0.10
$S + 2e \rightleftarrows S^{2-}$	−0.447
$Sn^{2+} + 2e \rightleftarrows Sn$	−0.1375
$Sn^{4+} + 2e \rightleftarrows Sn^{2+}$	0.15
$Tl^+ + e \rightleftarrows Tl$	−0.3363
$Tl^+ + e \rightleftarrows Tl(Hg)$	−0.3338
$Tl^{3+} + 2e \rightleftarrows Tl^+$	1.25
$U^{3+} + 3e \rightleftarrows U$	−1.66
$U^{4+} + e \rightleftarrows U^{3+}$	−0.52
$UO_2^+ + 4H^+ + e \rightleftarrows U^{4+} + 2H_2O$	0.273
$UO_2^{2+} + e \rightleftarrows UO_2^+$	0.163

(*continuec*